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Tsai

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(54) FOLDING TABLE

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(51) Int. Cl.

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A47B 3/091	(2006.01)
A47B 3/12	(2006.01)
A47B 3/10	(2006.01)

(52) U.S. Cl.
CPC ***A47B 3/06*** (2013.01); ***A47B 3/0918*** (2013.01); ***A47B 3/12*** (2013.01); *A47B 3/10* (2013.01)

(58) Field of Classification Search
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USPC 108/153.1, 163, 168
See application file for complete search history.

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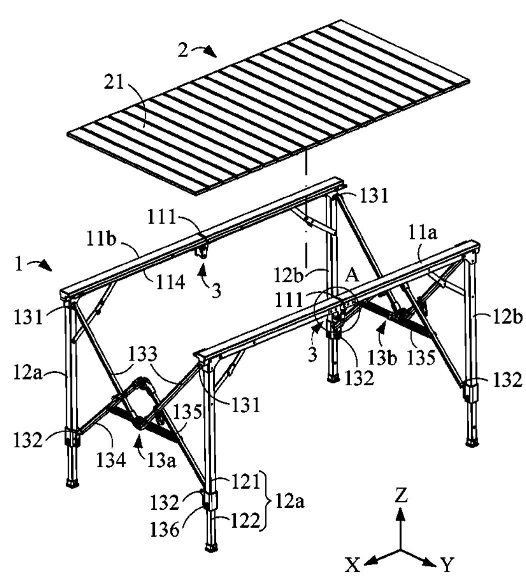
Primary Examiner — Daniel J Troy
Assistant Examiner — Timothy M Ayres

(57) ABSTRACT

A folding table includes a table frame, having two crossbars and a plurality of stands, the two crossbars each have a folding portion, the plurality of stands are respectively pivoted to two ends of the two crossbars; a table plate, detachably mounted on the two crossbars; and at least one gravity lock, disposed at the folding portion, the gravity lock has a body provided with two longitudinal long grooves, two side sheets are respectively located at two sides of the body, one end of the two side sheets is combined with the crossbar, and the other end is pivoted to the body, the two side sheets each have a hook portion to shade a partial opening of a slot.

6 Claims, 12 Drawing Sheets



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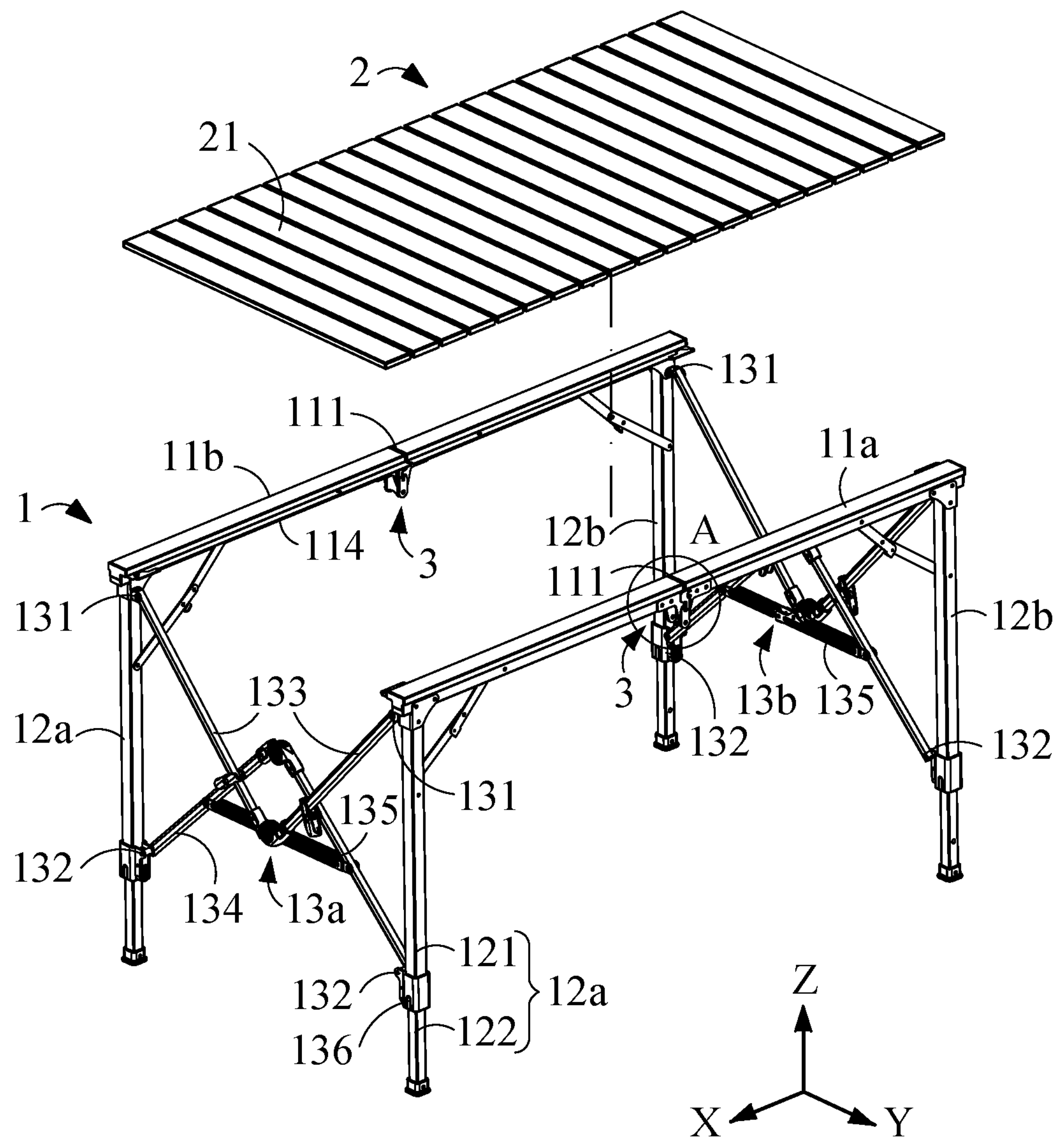
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114
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12b
A
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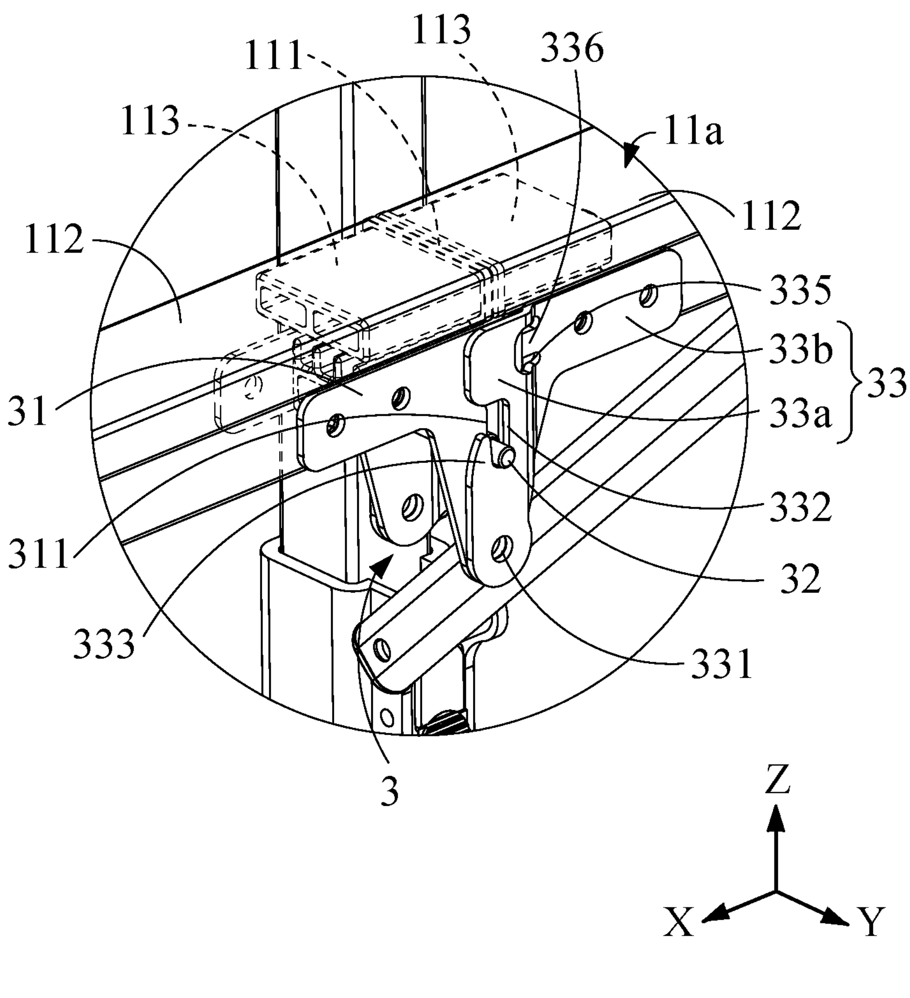
132
131
135
132
134
13a
121
12a
132
136
122
Z
X
Y

FIG. 1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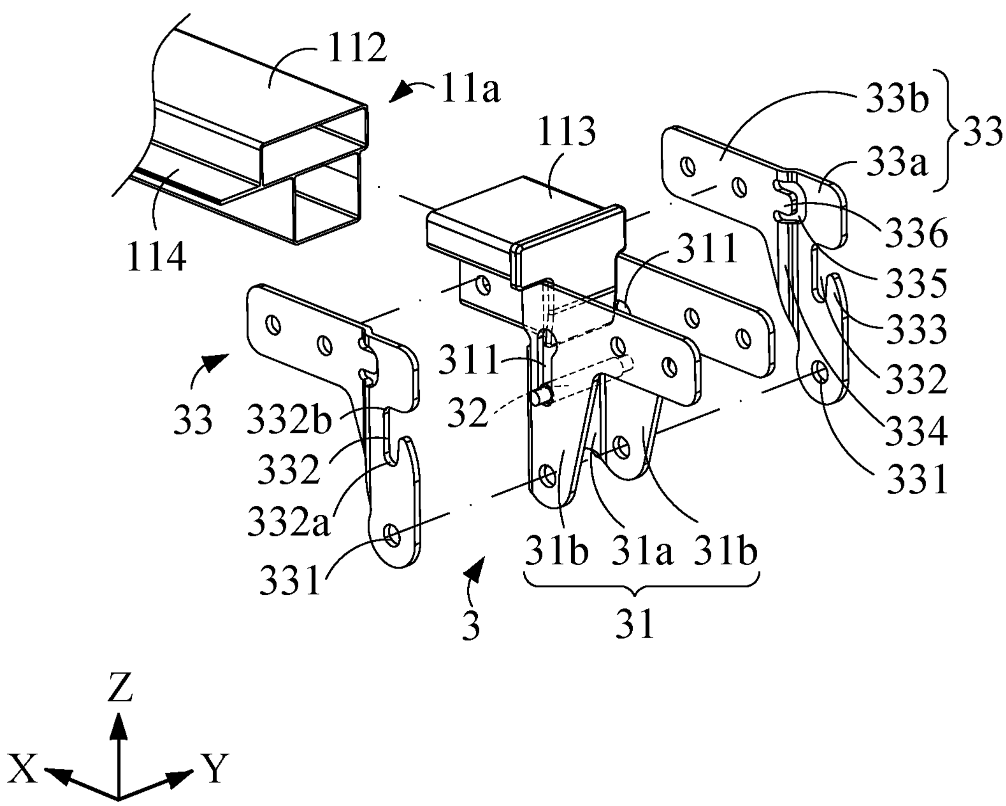
Y

FIG. 2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331
31b
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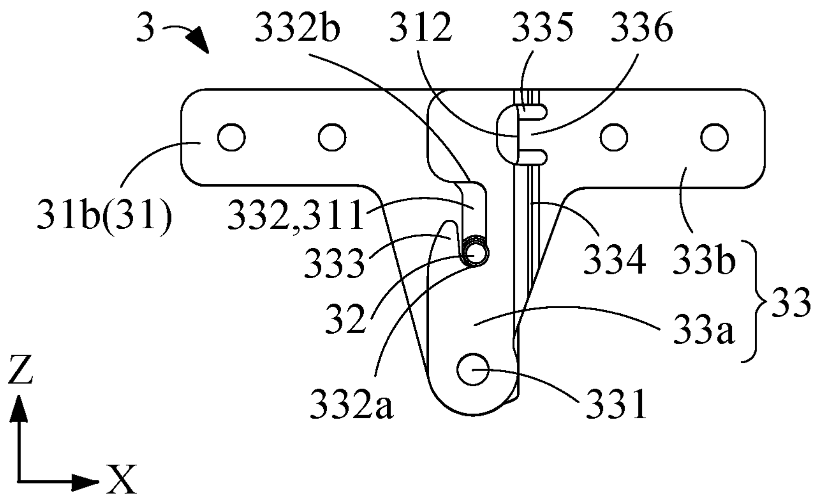
31b
3
31
Z
X
Y

FIG. 3


3
332b
312
335
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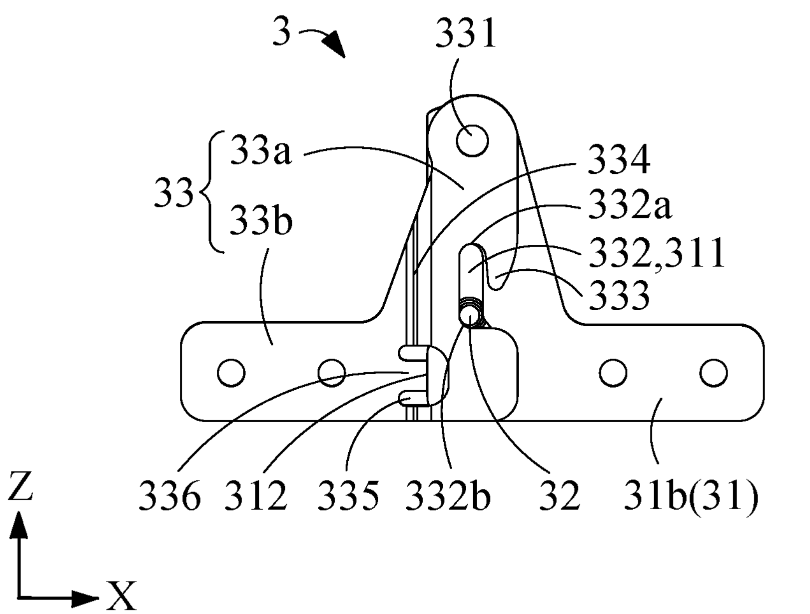
31b(31)
332,311
333
32
332a
334
33b
33
33a
331
Z
X

FIG. 4


3
331
33a
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332a
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336
312
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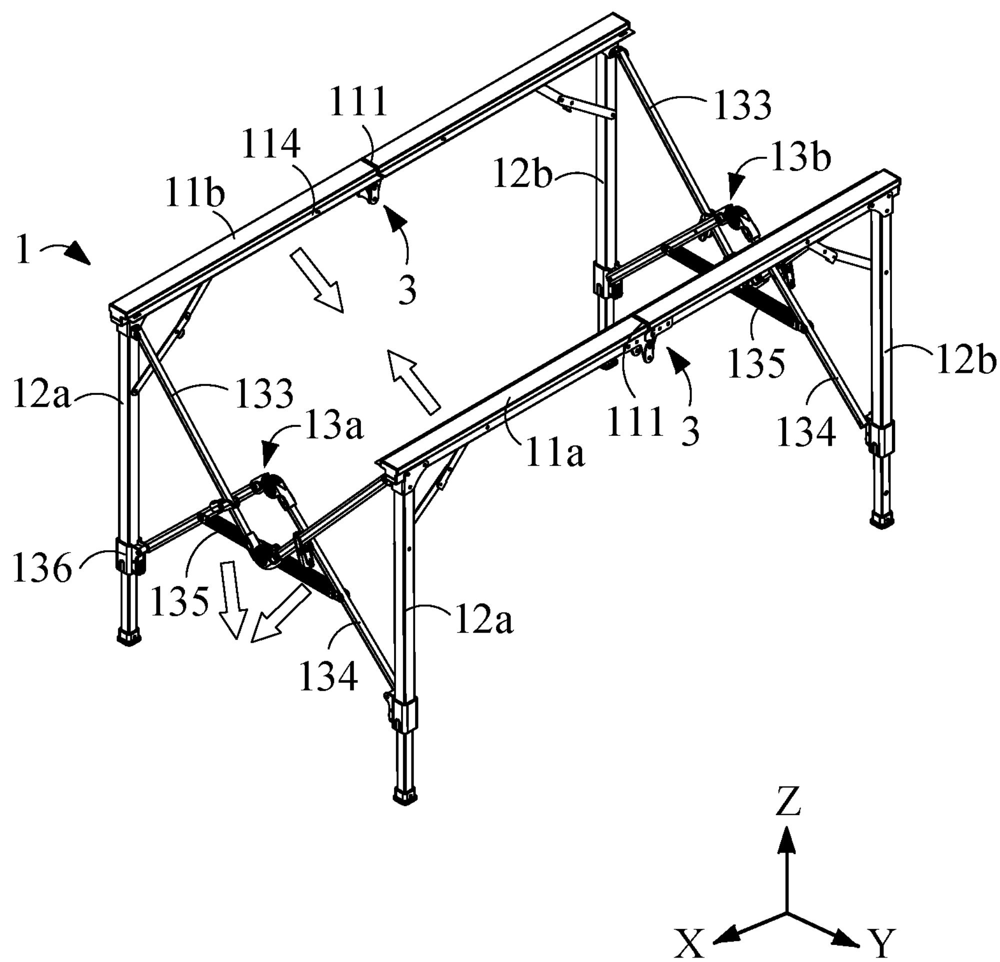
332b
32
31b(31)
Z
X

FIG. 5


111
114
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1
3
133
13b
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12a
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13a
111
3
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11a
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134
Z
X
Y

FIG. 6

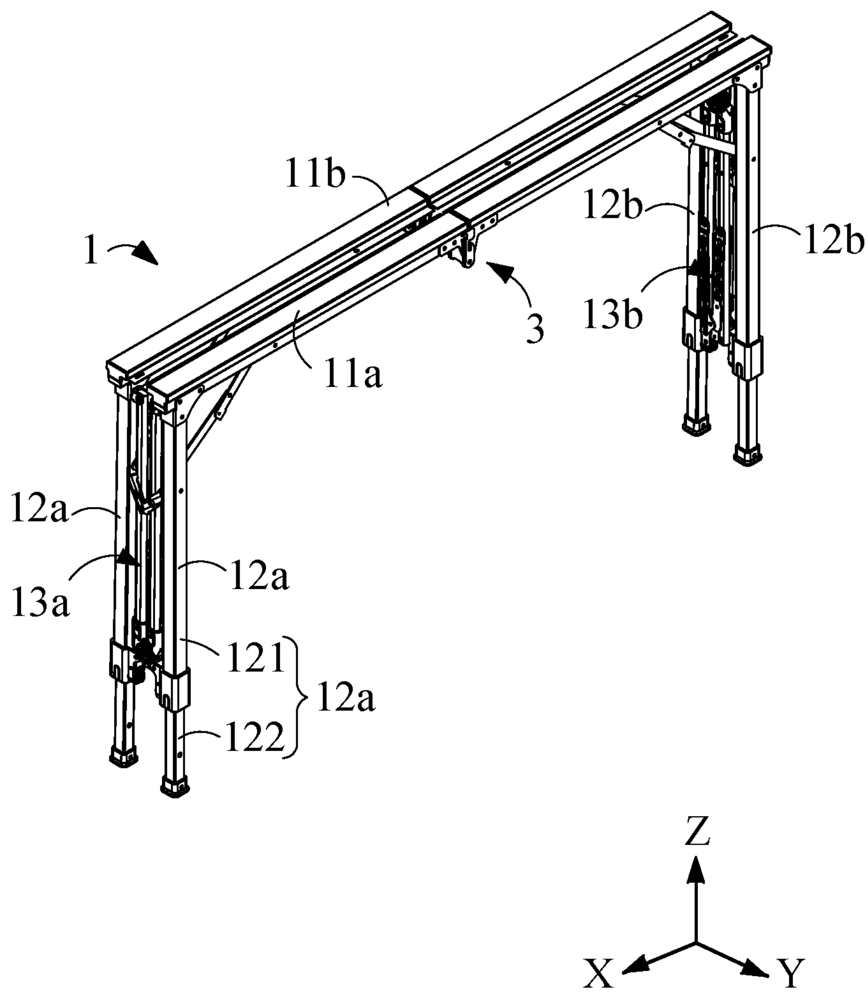

FIG. 7

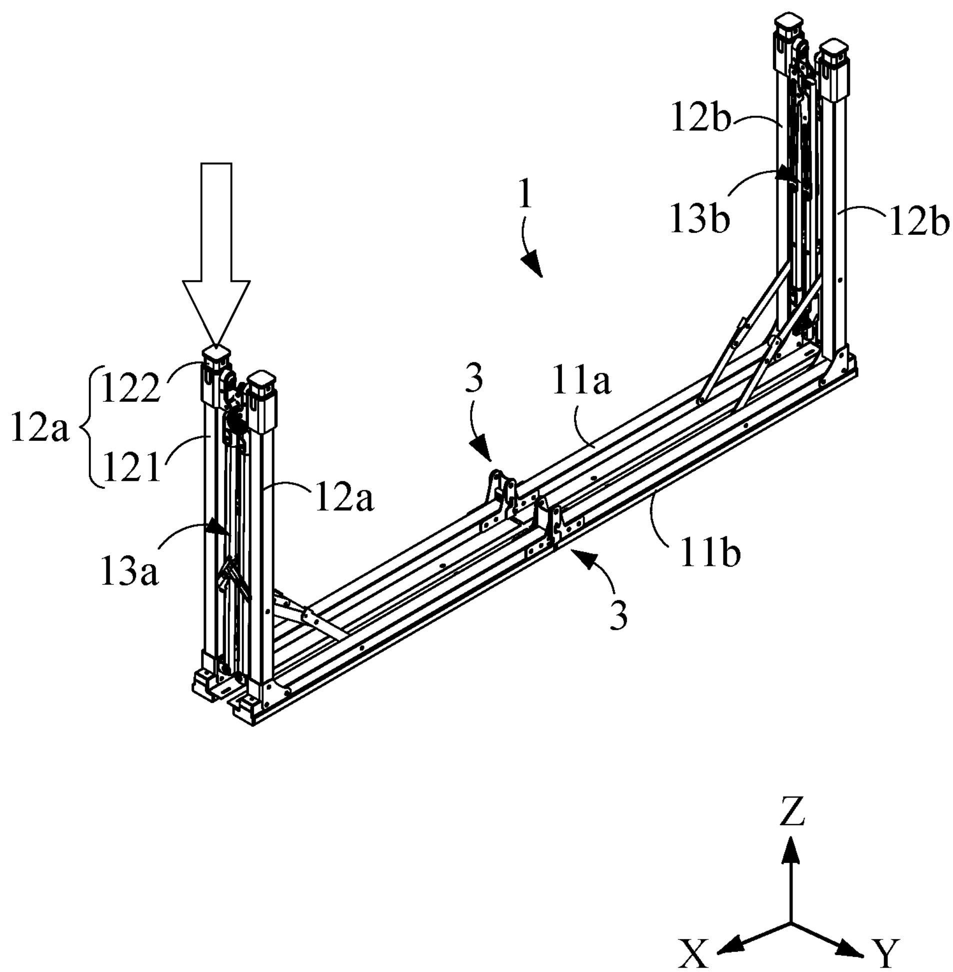

FIG. 8

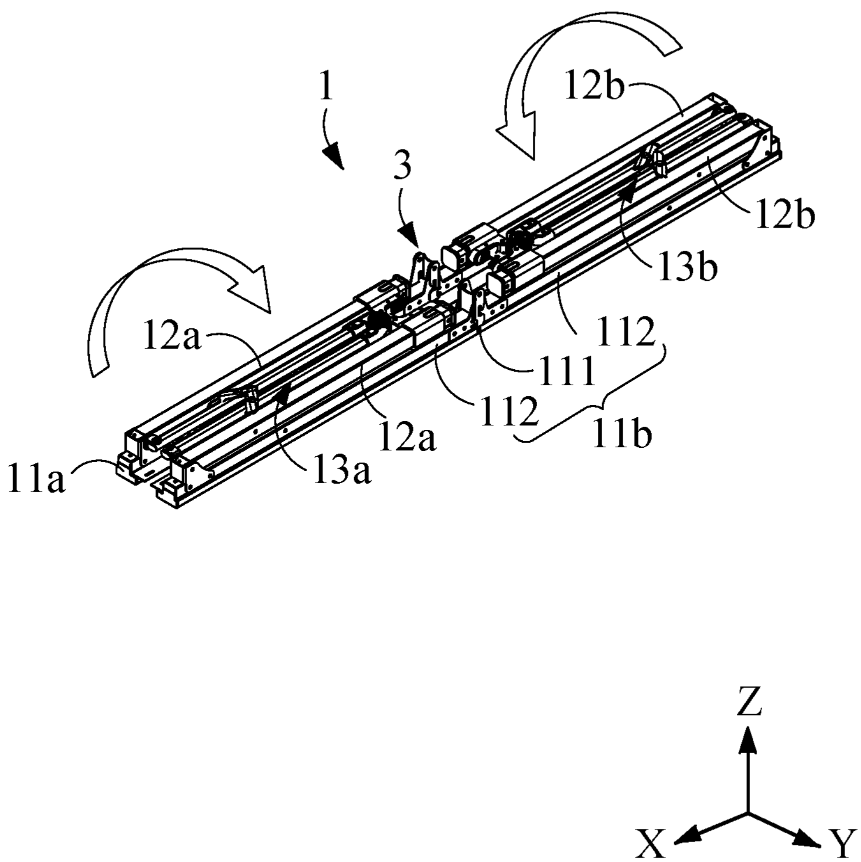

FIG. 9

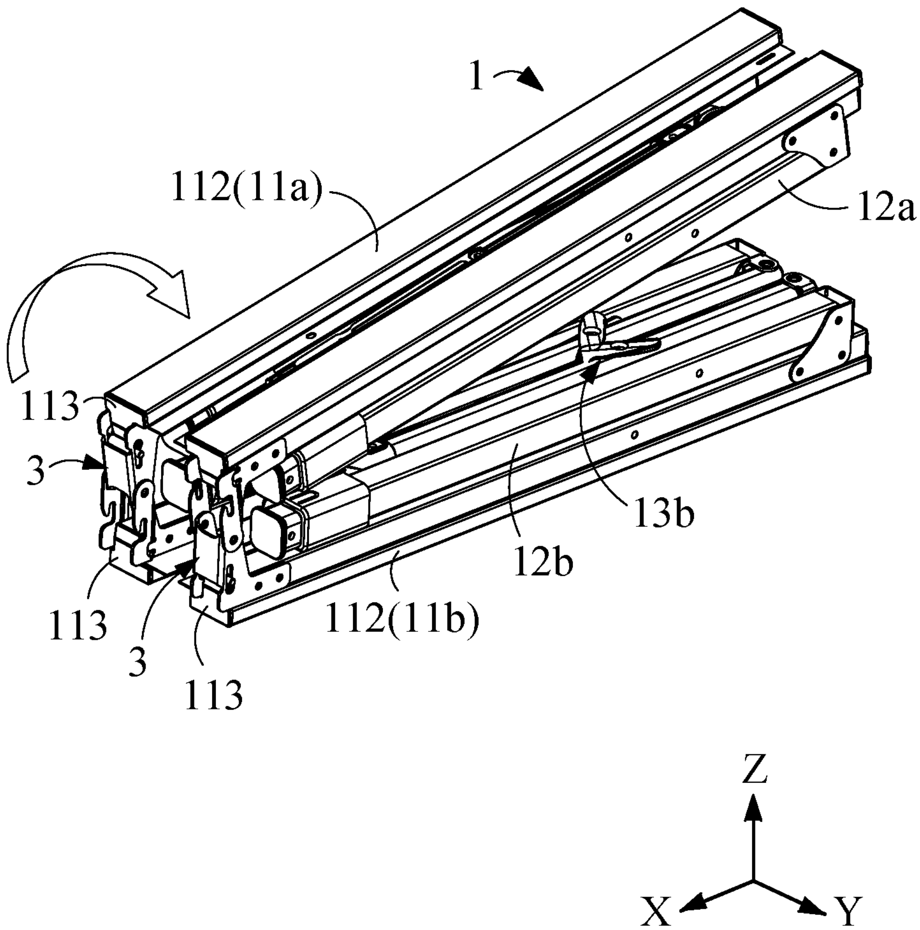

FIG. 10

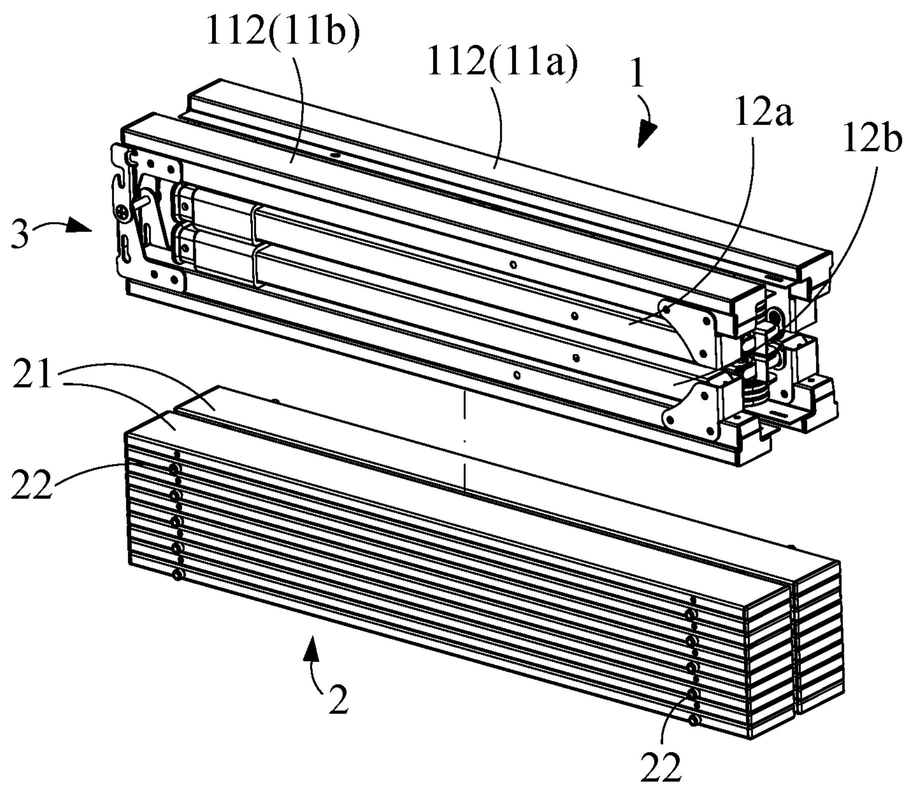

FIG. 11

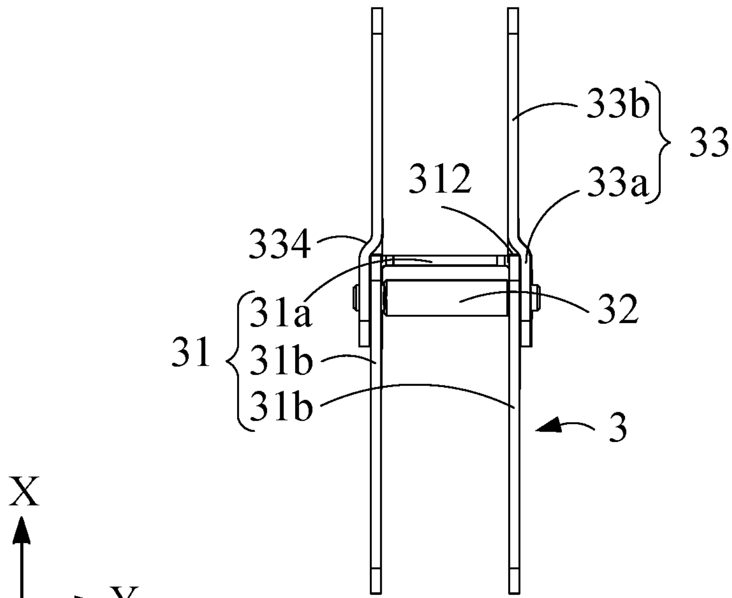

FIG. 12

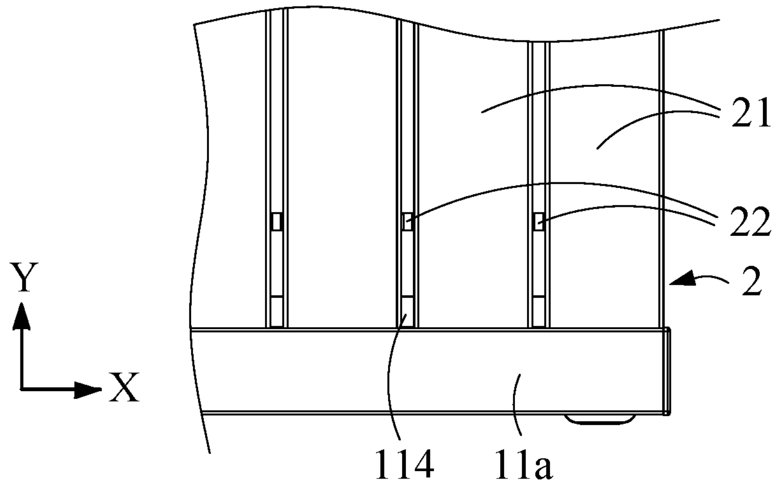

FIG. 13

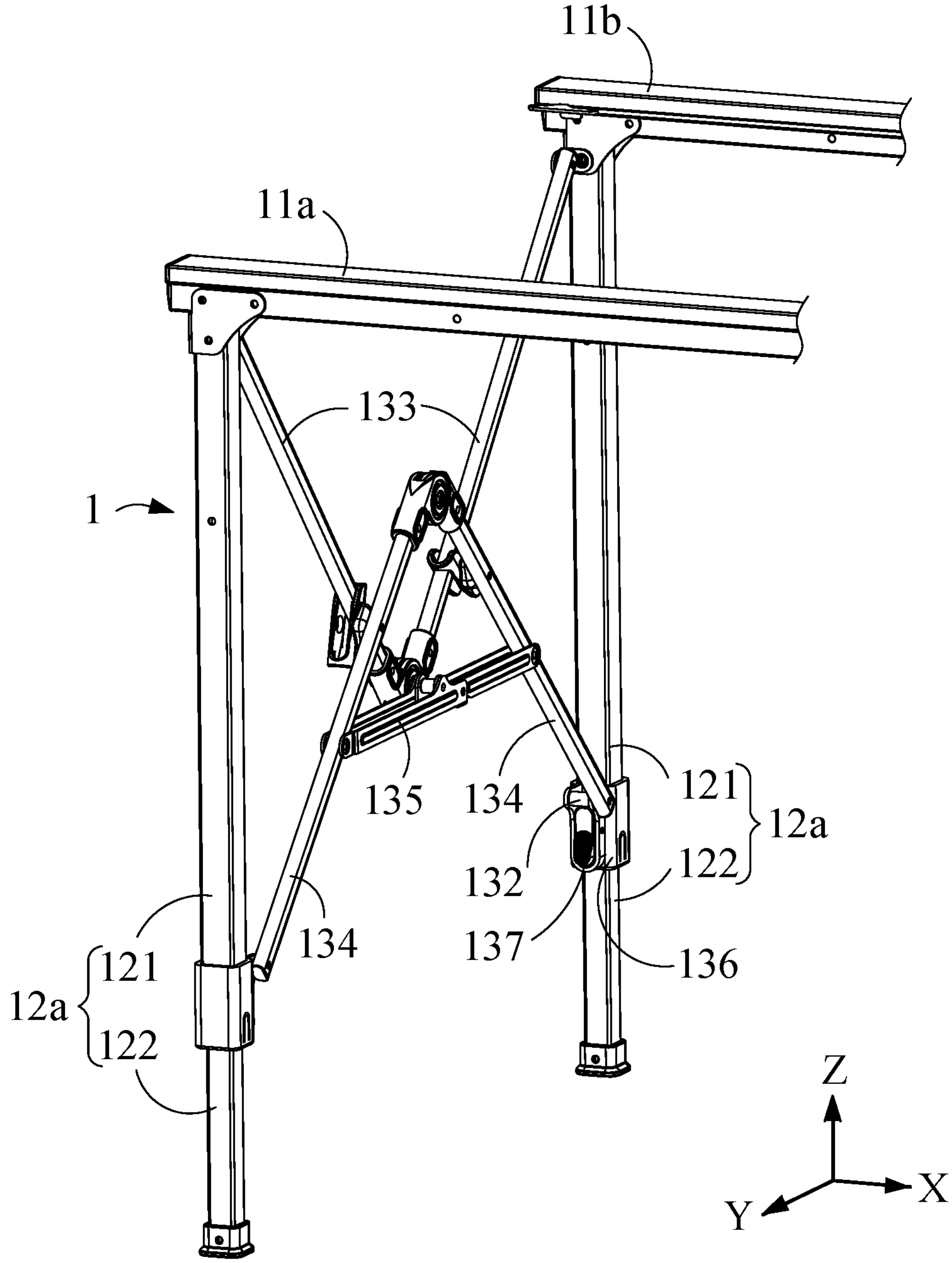

FIG. 14

FOLDING TABLE

BACKGROUND OF THE INVENTION

1. Field of the Invention

The present disclosure relates to a table, in particular to a folding table that can be folded into a small volume for storage or carrying.

2. Description of the Related Art

The table plate and table legs of the common traditional folding table are fixed and non-detachable, but the table plate can be retracted to one side of the table legs by pivoting relative to the table legs, and the unfolded table legs can also be folded in the same direction, so that when the folding table is switched from an unfolded state to a folded state, the thickness in a single direction can be reduced, so as to reduce the storage space occupied.

BRIEF SUMMARY OF THE INVENTION

However, even if the traditional folding table is folded, the overall volume of the folding table is still too large to place into a narrow space (such as the rear compartment of a car), thus limiting the convenience of the user when storing or moving such tables; for occasions where the table needs to be carried frequently and used outdoors, such as outdoor banquets or camping, the traditional folding table is not suitable.

Furthermore, when the traditional folding table is switched to the unfolded or folded state, additional operations are required to lock or unlock the locking mechanism, so the operating convenience and efficiency are not good.

In view of the above prior art, the disclosure provides a folding table, wherein the table plate and the table frame are detachable, and at least the table frame can be folded into a small volume; in addition, the locking mechanism of the folding table can be automatically locked or unlocked according to the use state.

The directionality or similar terms thereof described in the full text of the present disclosure, such as “front”, “rear”, “left”, “right”, “upper (top)”, “lower (bottom)”, “inner”, “outer”, “side”, etc., are mainly with reference to the direction of the attached drawings, and each directionality or similar terms thereof are only used to assist in explaining and understanding the embodiments of the present disclosure, and are not used to limit the present disclosure.

The use of the quantifier “a” or “one” in the elements and components described in the full text of the present disclosure is only for convenience of use and to provide a common meaning of the scope of the present disclosure; in the present disclosure, it should be construed to include one or at least one, and a singular concept also includes the plural case, unless it is expressly intended to mean something else.

The similar terms such as “combine”, “fit together” or “assemble” mentioned in the full text of the present disclosure mainly include the forms in which the components can be separated without destroying after being connected, or the components are inseparable after being connected, etc., which can be selected by a person having ordinary knowledge in the art according to the material or assembly requirements of the components to be connected.

To achieve the above objective and other objectives, the present disclosure provides a folding table, including: a table frame, having a first crossbar, a second crossbar, two first side stands and two second side stands, the first crossbar and the second crossbar each have a folding portion, the two first side stands are respectively pivoted to first ends of the first crossbar and the second crossbar, and two second side stands are respectively pivoted to second ends of the first crossbar and the second crossbar, a first side support set is respectively pivoted to the two first side stands by two upper junctions and two lower junctions; a second side support set is respectively pivoted to the two second side stands by two upper junctions and two lower junctions; a table plate, when the table frame is in an unfolded state, the table plate is detachably mounted on the first crossbar and the second crossbar; and at least one gravity lock, disposed at the folding portion of the first crossbar and/or the folding portion of the second crossbar, the gravity lock has a body, a lock lever and two side sheets, the body is provided with two longitudinal long grooves, the two side sheets are respectively located at two sides of the body, one end of the two side sheets is combined with the first crossbar or the second crossbar, and the other end is pivoted to the body by a pivot portion, the two side sheets are each provided with a slot, and a first end of the slot is closer to the pivot portion than a second end, the two side sheets each have a hook portion extending from the first end to the second end of the slot to shade a partial opening of the slot; in a locked state, the slots of the two side sheets are respectively aligned with the two longitudinal long grooves, the lock lever is penetratingly disposed in the two longitudinal long grooves and the slots of the two side sheets, the lock lever is located within the scope of the hook portion, so that the lock lever is clamped by the hook portion, and the two side sheets are limited from pivoting relative to the body; when the gravity lock is inverted, the lock lever can be detached from the scope of the hook portion and switched to an unlocked state of the gravity lock, so that the two side sheets can pivot relative to the body.

In the above folding table, the side sheet of the gravity lock may have a clamping plate and a locking plate, the pivot portion, the slot and the hook portion are all located at the clamping plate, and the side sheet is combined to the first crossbar or the second crossbar by the locking plate, the clamping plate and the locking plate can be connected by an inclined portion; the body can have an intermediate sheet connecting two opposite sheets, the two longitudinal long grooves may be respectively disposed at the two opposite sheets, in the locked state of the gravity lock, the inclined portions of the two side sheets can be aligned at rear end faces of the adjacent opposite sheets.

In the above folding table, the side sheet may have a hollow out portion, extending from the locking plate to the clamping plate through the inclined portion, the locking plate may have an abutting piece protruding in the hollow out portion, in the locked state of the gravity lock, the abutting piece can be abutted to the rear end face of the opposite sheet.

In the above folding table, the first crossbar has two segmented pipes and two end plugs, and the two end plugs are respectively combined with one end of the two segmented pipes, the locking plate of the two side sheets can be combined with one segmented pipe and end plug thereof, and the two opposite sheets can be combined with another segmented pipe and end plug thereof; in the unfolded state of the table frame, the two end plugs can be abutted against each other and form the folding part of the first crossbar.

In the above folding table, the first crossbar and the second crossbar each have an inner flange, and the table plate is detachably disposed at the two inner flanges.

In the above folding table, the table plate has a plurality of unit plates, the plurality of unit plates are placed side by side at the two inner flanges, and any two adjacent unit plates are abutted by at least one spacer pad.

In the above folding table, the first side support set may have two upper rods and two lower rods, one end of the two upper rods is pivoted and combined with a horizontal locking sheet, and the other ends respectively form the two upper junctions; one end of the two lower rods is pivoted, and the other ends respectively form the two lower junctions, and two ends of the horizontal locking sheet are pivoted to the two lower rods.

In the above folding table, the two first side stands may each have an outer pipe and an inner pipe that are partially sleeved, the first side support set may have two pipe sleeves, the two pipe sleeves are respectively sleeved on the two outer pipes, and for pivoting the two lower junctions, the first side support set may have two elastic locking members respectively disposed at the two pipe sleeves, when the elastic locking member is not pressed, the inner pipe can be limited from stretching and contracting relative to the outer pipe.

Accordingly, the folding table of the present disclosure has the table plate and the table frame that are detachable, in the process of switching the table frame from the unfolded state to the folded state, it has undergone a volume reduction of multiple dimensions, so that the volume of the overall table frame can be greatly reduced, so that it can be easily placed in a narrow space (such as the rear compartment of a car), and the convenience of storage or movement is improved, so it is very suitable for being used in the occasion that the table needs to be carried frequently and used outdoors. In addition, in the process of switching to the unfolded or folded state of the table frame, the gravity lock can be automatically locked or unlocked, so that it can further improve the convenience and efficiency of operation, as well as the safety of use.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a schematic perspective exploded view of an embodiment of the present disclosure.

FIG. **2** is an enlarged view of area A in FIG. **1**.

FIG. **3** is a schematic perspective exploded view of a gravity lock and a partial table frame of an embodiment of the present disclosure.

FIG. **4** is a side view of the gravity lock being in a locked state of an embodiment of the present disclosure.

FIG. **5** is a side view of the gravity lock being in an unlocked state of an embodiment of the present disclosure.

FIG. **6** is a schematic view of folding action of the table frame from an unfolded state of an embodiment of the present disclosure.

FIG. **7** is a schematic perspective view of a first crossbar and a second crossbar of the table frame being brought together of an embodiment of the present disclosure.

FIG. **8** is a schematic perspective view of a folding table in the inverted state of FIG. **7**.

FIG. **9** is a schematic view of the action when folding the stands.

FIG. **10** is a schematic view of the action when folding the first crossbar and the second crossbar.

FIG. **11** is a schematic perspective view of both the table frame and the table plate being in the folded state of an embodiment of the present disclosure.

FIG. **12** is a top view of the gravity lock being in a locked state of an embodiment of the present disclosure.

FIG. **13** is a schematic partial top view of the table frame of an embodiment of the present disclosure.

FIG. **14** is schematic perspective view of the table frame being an unfolded state from another viewing angle of an embodiment of the present disclosure.

DETAILED DESCRIPTION OF THE INVENTION

To facilitate understanding of the purpose, characteristics and effects of the present disclosure, embodiments together with the attached drawings for the detailed description of the present disclosure are provided as below.

Referring to FIG. **1**, FIG. **1** is a preferable embodiment of a folding table of the present disclosure, the folding table includes a table frame **1**, a table plate **2** and at least one gravity lock **3**, the table plate **2** is detachably disposed at the table frame **1**, and the gravity lock **3** is disposed at the table frame **1**.

In detail, the table frame **1** has a first crossbar **11***a* and a second crossbar **11***b* spaced apart in a Y direction in an unfolded state, the first crossbar **11***a* and the second crossbar **11***b* can each extend along a X direction, and the first crossbar **11***a* and the second crossbar **11***b* each have a folding portion **111**. Two first side stands **12***a* are respectively pivoted to first ends of the first crossbar **11***a* and the second crossbar **11***b*, and two second side stands **12***b* are respectively pivoted to second ends of the first crossbar **11***a* and the second crossbar **11***b*; the two first side stands **12***a* and the two second side stands **12***b* can each extend along a Z direction. The table frame **1** also has a first side support set **13***a* and a second side support set **13***b* spaced apart in the X direction, and the first side support set **13***a* is respectively pivoted to the two first side stands **12***a* by two upper junctions **131** and two lower junctions **132**; the second side support set **13***b* is respectively pivoted to the two second side stands **12***b* by two upper junctions **131** and two lower junctions **132**.

The table plate **2** is detachably mounted on the first crossbar **11***a* and the second crossbar **11***b*, and covers a space between the first crossbar **11***a* and the second crossbar **11***b*. The gravity lock **3** is disposed at the folding portion **111** of the first crossbar **11***a* and/or the folding portion **111** of the second crossbar **11***b*, so that the first crossbar **11***a* and the second crossbar **11***b* are controlled not to be folded when locked, but can be folded when unlocked.

Referring to FIG. **2** to FIG. **5**, the gravity lock **3** has a body **31**, a lock lever **32** and two side sheets **33**. The body **31** is provided with two longitudinal long grooves **311** spaced apart in the Y direction, and the two longitudinal long grooves **311** each extend along the Z direction. The lock lever **32** is penetratingly disposed in the two longitudinal long grooves **311**, and two ends of the lock lever **32** respectively penetrate and extend out of the corresponding longitudinal long grooves **311**. The two side sheets **33** are respectively located at two sides of the body **31** in the Y direction, one end of the two side sheets **33** is combined with the first crossbar **11***a* or the second crossbar **11***b*, and the other end is pivoted to the body **31** by a pivot portion **331**. The two side sheets **33** are each provided with a slot **332**, and a first end **332***a* of the slot **332** is closer to the pivot portion **331** than a second end **332***b*; the two side sheets **33** each have a hook portion **333** extending from the first end **332***a* to the second end **332***b* of the slot **332**, so as to shade a partial opening of the slot **332** by the hook portion **333**.

The gravity lock **3** is in the locked state as shown in FIG. **4**, the slots **332** of the two side sheets **33** are respectively aligned with the two longitudinal long grooves **311** of the body **31**, and the two ends of the lock lever **32** can respectively penetrate the slots **332** of the two side sheets **33**; at this time, the lock lever **32** is located at the first end **332***a* of the slot **332** by gravity, and is located within the scope of the hook portion **333**, so that the lock lever **32** is clamped by the hook portion **333**, and the two side sheets **33** are limited from pivoting relative to the body **31**.

The gravity lock **3** can be inverted to switch the gravity lock **3** to the unlocked state as shown in FIG. **5**. That is, at this time, the lock lever **32** can be affected by gravity and fall towards the second end **332***b* of the slot **332**, thereby detaching from the scope of the hook portion **333**, so that the two side sheets **33** can pivot relative to the body **31**.

Referring to FIG. **1** and FIG. **2**, in the unfolded state of the folding table, because the two side sheets **33** of the gravity lock **3** are facing downwards with the first end **332***a* of the slot **332**, the gravity lock **3** can automatically appear locked, so that the folding parts **111** of the first crossbar **11***a* and the second crossbar **11***b* are limited by the gravity lock **3** and cannot be folded, and the unfold state can be remained in a straight line, so that the situation that the folding table is accidentally deformed in use can be avoided.

Referring to FIG. **1**, FIG. **6** and FIG. **7**, to fold the folding table, the table plate **2** may be detached from the first crossbar **11***a* and the second crossbar **11***b*. Then, as shown in FIG. **6**, the first side support set **13***a* and the second side support set **13***b* are folded, so that the first crossbar **11***a* and the second crossbar **11***b* can be brought together to form the state shown in FIG. **7**, and the volume of the folding table in the Y direction is reduced. At this time, the gravity lock **3** remains in the locked state.

Referring to FIG. **8**, continuously, the folding table is inverted, the first crossbar **11***a* and the second crossbar **11***b* are placed on the ground, and the two first side stands **12***a* and the two second side stands **12***b* are facing upwards. At this time, referring to FIG. **5** together, the gravity lock **3** can be automatically switched to the unlocked state because of inversion.

Referring to FIG. **9**, continuously, the second first side post **12***a* and the two second side stands **12***b* are folded inward to reduce the volume of the folding table in the Z direction.

Referring to FIG. **5** and FIG. **10**, because the gravity lock **3** has been switched to the unlocked state, the first crossbar **11***a* and the second crossbar **11***b* can be folded successively according to their respective folding portions **111**, so as to reduce the volume of the folding table in the X direction, so that the table frame **1** can be converted into a folded state as shown in FIG. **11**.

Conversely, to switch the table frame **1** from the folded state back to the unfolded state, the user only needs to reverse the preceding steps. Among them, when the table frame **1** changes to the state shown in FIG. **7**, the gravity lock **3** can automatically switch back to the locked state.

Therefore, regarding the folding table of the present embodiment, in the process of switching the table frame **1** from the unfolded state to the folded state, it has undergone a volume reduction of three dimensions, so that the volume of the overall table frame **1** can be greatly reduced, so that it can be easily placed in a narrow space (such as the rear compartment of a car), and the convenience of storage or movement is improved, so it is very suitable for being used in the occasion that the table needs to be carried frequently and used outdoors. In addition, in the process of switching to the unfolded or folded state of the table frame **1**, the gravity lock **3** can be automatically locked or unlocked, so that it can further improve the convenience and efficiency of operation, as well as the safety of use.

Referring to FIG. **2** to FIG. **4** and FIG. **12**, in addition to the above embodiments, in one embodiment of the present disclosure, the side sheet **33** of the gravity lock **3** may have a clamping plate **33***a* and a locking plate **33***b*, the pivot portion **331**, the slot **332** and the hook portion **333** are all located at the clamping plate **33***a*, and the side sheet **33** is combined to the first crossbar **11***a* or the second crossbar **11***b* by the locking plate **33***b*; the clamping plate **33***a* and the locking plate **33***b* can be connected by an inclined portion **334**, and the side sheet **33** may be approximately L-shaped. The body **31** has an intermediate sheet **31***a* connecting two opposite sheets **31***b*, and the two opposite sheets **31***b* can be opposite in the Y direction, so that the cross-section of the lower half of the body **31** on the XY plane may be approximately U-shaped, and the two longitudinal long grooves **311** may be respectively disposed at the two opposite sheets **31***b*. In the locked state of the gravity lock **3**, the clamping plates **33***a* of the two side sheets **33** can be respectively located at the opposite outer sides of the two opposite sheets **31***b*, and the slots **332** of the two side sheets **33** can be respectively aligned at the two longitudinal long grooves **311**, and the inclined portions **334** of the two side sheets **33** can be aligned at rear end faces **312** of the adjacent opposite sheets **31***b*. In this way, the gravity lock **3** of the present embodiment can be firmly combined to the two sides of the first crossbar **11***a* or the second crossbar **11***b* by the two side sheets **33** and the two opposite sheets **31***b* of the body **31** respectively, and has the effect of improving the convenience of assembly.

In addition to the above embodiments, in one embodiment of the present disclosure, the side sheet **33** may have a hollow out portion **335**, the hollow out portion **335** may extend from the locking plate **33***b* to the clamping plate **33***a* through the inclined portion **334**, and the locking plate **33***b* may also have an abutting piece **336** protruding in the hollow out portion **335**. In the locked state of the gravity lock **3**, the abutting piece **336** can be abutted to the rear end face **312** of the opposite sheet **31***b*. In this way, the gravity lock **3** of the present embodiment can limit the approaching position of the locking plate **33***b* and the opposite sheet **31***b* by a simple structure that is easy to form, so that the corresponding slot **332** and the longitudinal long groove **311** can be accurately opposed, the locking lever **32** is remained without being clamped and can be smoothly switched in the groove by the action of gravity.

Referring to FIG. **2** and FIG. **3**, in addition to the above embodiments, in one embodiment of the present disclosure, the first crossbar **11***a* has two segmented pipes **112** and two end plugs **113**, and the two end plugs **113** are respectively combined with one end of the two segmented pipes **112**. The locking plate **33***b* of the two side sheets **33** is combined with one segmented pipe **112** and end plug **113** thereof, and the two opposite sheets **31***b* are combined with another segmented pipe **112** and end plug **113** thereof; in the unfolded state of the table frame **1**, the two end plugs **113** are abutted against each other and form the folding part **111** of the first crossbar **11***a*. In this way, the embodiment can achieve that the first crossbar **11***a* is convenient for being unfolded or folded in the X direction by a simple structure. The second crossbar **11***b* has the same structure as the first crossbar **11***a* with two segmented pipes **112** and two end plugs **113**, so they are not described in detail herein.

Referring to FIG. **1** and FIG. **3**, in addition to the above embodiments, in one embodiment of the present disclosure, the first crossbar **11***a* and the second crossbar **11***b* each have an inner flange **114**, and the table plate **2** is detachably disposed at the two inner flanges **114**. In this way, the table frame **1** of the present embodiment can be used for firmly mounting the table plate **2** by a simple structure and is convenient for disassembling or assembling, and has the effects such as improving the operation convenience and efficiency when disassembling or assembling.

Referring to FIG. **1** and FIG. **13**, in addition to the above embodiment, in one of the embodiments of the present disclosure, the table plate **2** has a plurality of unit plates **21**, the plurality of unit plates **21** are placed side by side at the two inner flanges **114**, and any two adjacent unit plates **21** are abutted by at least one spacer pad **22**. In this way, the embodiment can provide the user to quickly place the plurality of unit plates **21** side by side at the two inner flanges **114** at equal intervals to form the table board **2**, and the plurality of unit plates **21** can be conveniently disassembled and stored and stacked into a small volume as shown in FIG. **11**, so as to improve the convenience when storing or moving.

Referring to FIG. **1** and FIG. **14**, in addition to the above embodiments, in one embodiment of the present disclosure, the first side support set **13***a* may have two upper rods **133** and two lower rods **134**, one end of the two upper rods **133** is pivoted and combined with a horizontal locking sheet **135**, and the other ends respectively form the two upper junctions **131**; one end of the two lower rods **134** is pivoted, and the other ends respectively form the two lower junctions **132**, and two ends of the horizontal locking sheet **135** are pivoted to the two lower rods **134**. In this way, the first side support set **13***a* of the present embodiment can remain the opening state of the two lower rods **134** and simultaneously limit the opening state of the two upper rods **133** when the horizontal locking sheet **135** is unfolded into a straight line; during the folding step as shown in FIG. **6**, the horizontal locking sheet **135** can be pushed down and bent, so that the first side support set **13***a* can be folded. The second side support set **13***b* has the same structure as the first side support set **13***a* with the two upper rods **133**, the two lower rods **134** and the horizontal locking sheet **135**, they are not described in detail herein.

In addition to the above implementation, in one embodiment of the present disclosure, the two first side stands **12***a* may each have an outer pipe **121** and an inner pipe **122** that are partially sleeved; the first side support set **13***a* may have two pipe sleeves **136**, the two pipe sleeves **136** are respectively sleeved on the two outer pipes **121**, and for pivoting the two lower junctions **132**, the first side support set **13***a* may have two elastic locking members **137**, the two elastic locking members **137** are respectively disposed at the two pipe sleeves **136**, when the elastic locking member **137** is not pressed, the inner pipe **122** can be limited from stretching and contracting relative to the outer pipe **121**; when the length of the first side stand **12***a* is to be adjusted, the elastic locking member **137** can be pressed, so that the inner pipe **122** can be stretched and contracted relative to the outer pipe **121**. During the folding step as shown in FIG. **8**, the elastic locking member **137** can be pressed first, and the first side stand **12***a* is reduced to shortest, and then folded toward the first crossbar **11***a* or the second crossbar **11***b*. The two second side stands **12***b* can also have an outer pipe **121** and an inner pipe **122** that are partially sleeved, and the second side support set **13***b* has the same structure as the first side support set **13***a* with two pipe sleeves **136** and two elastic locking members **137**, so they are not described in detail herein. In this way, the embodiment can facilitate the adjustment of the length of the two first side stands **12***a* and the two second side stands **12***b*.

While the present invention has been described by means of preferable embodiments, those skilled in the art should understand the above description is merely for describing the invention, and it should not be considered to limit the scope of the invention. It should be noted that all changes and substitutions which come within the meaning and range of equivalency of the embodiments are intended to be embraced in the scope of the invention. Therefore, the scope of the invention is defined by the claims.

What is claimed is:

1. A folding table, comprising:

a table frame, having a first crossbar, a second crossbar, two first side stands and two second side stands, the first crossbar and the second crossbar each have a folding portion, the two first side stands are respectively pivoted to first ends of the first crossbar and the second crossbar, and two second side stands are respectively pivoted to second ends of the first crossbar and the second crossbar, a first side support set is respectively pivoted to the two first side stands by two upper junctions and two lower junctions; a second side support set is respectively pivoted to the two second side stands by two upper junctions and two lower junctions;

a table plate, when the table frame is in an unfolded state, the table plate is detachably mounted on the first crossbar and the second crossbar; and

at least one gravity lock, disposed at the folding portion of the first crossbar and/or the folding portion of the second crossbar, the gravity lock has a body, a lock lever and two side sheets, the body is provided with two longitudinal long grooves, the two side sheets are respectively located at two sides of the body, one end of the two side sheets is combined with the first crossbar or the second crossbar, and the other end is pivoted to the body by a pivot portion, the two side sheets are each provided with a slot, and a first end of the slot is closer to the pivot portion than a second end, the two side sheets each have a hook portion extending from the first end to the second end of the slot to shade a partial opening of the slot; in a locked state, the slots of the two side sheets are respectively aligned with the two longitudinal long grooves, the lock lever is penetratingly disposed in the two longitudinal long grooves and the slots of the two side sheets, the lock lever is located within the scope of the hook portion, so that the lock lever is clamped by the hook portion, and the two side sheets are limited from pivoting relative to the body; when the gravity lock is inverted, the lock lever can be detached from the scope of the hook portion and switched to an unlocked state of the gravity lock, so that the two side sheets can pivot relative to the body,

wherein the side sheet of the gravity lock has a clamping plate and a locking plate, the pivot portion, the slot and the hook portion are all located at the clamping plate, and the side sheet is combined to the first crossbar or the second crossbar by the locking plate, the clamping plate and the locking plate are connected by an inclined portion; the body has an intermediate sheet connecting two opposite sheets, the two longitudinal long grooves are respectively disposed at the two opposite sheets, in the locked state of the gravity lock, the inclined portions of the two side sheets are aligned at rear end faces of the adjacent opposite sheets, and wherein the side sheet has a hollow out portion, extending from the locking plate to the clamping plate through the inclined portion, the locking plate has an abutting piece protruding in the hollow out portion, in the locked state of the gravity lock, the abutting piece is abutted to the rear end face of the opposite sheet.

2. The folding table according to claim **1**, wherein the first crossbar and the second crossbar each have an inner flange, and the table plate is detachably disposed at the two inner flanges.

3. The folding table according to claim **2**, wherein the table plate has a plurality of unit plates, the plurality of unit plates are placed side by side at the two inner flanges, and any two adjacent unit plates are abutted by at least one spacer pad.

4. The folding table according to claim **1**, wherein the first side support set has two upper rods and two lower rods, one end of the two upper rods is pivoted and combined with a horizontal locking sheet, and the other ends respectively form the two upper junctions; one end of the two lower rods is pivoted, and the other ends respectively form the two lower junctions, and two ends of the horizontal locking sheet are pivoted to the two lower rods.

5. The folding table according to claim **4**, wherein the two first side stands each have an outer pipe and an inner pipe that are partially sleeved, the first side support set has two pipe sleeves, the two pipe sleeves are respectively sleeved on the two outer pipes, and for pivoting the two lower junctions, the first side support set has two elastic locking members respectively disposed at the two pipe sleeves, when the elastic locking member is not pressed, the inner pipe can be limited from stretching and contracting relative to the outer pipe.

6. A folding table, comprising:

a table frame, having a first crossbar, a second crossbar, two first side stands and two second side stands, the first crossbar and the second crossbar each have a folding portion, the two first side stands are respectively pivoted to first ends of the first crossbar and the second crossbar, and two second side stands are respectively pivoted to second ends of the first crossbar and the second crossbar, a first side support set is respectively pivoted to the two first side stands by two upper junctions and two lower junctions; a second side support set is respectively pivoted to the two second side stands by two upper junctions and two lower junctions;

a table plate, when the table frame is in an unfolded state, the table plate is detachably mounted on the first crossbar and the second crossbar; and

at least one gravity lock, disposed at the folding portion of the first crossbar and/or the folding portion of the second crossbar, the gravity lock has a body, a lock lever and two side sheets, the body is provided with two longitudinal long grooves, the two side sheets are respectively located at two sides of the body, one end of the two side sheets is combined with the first crossbar or the second crossbar, and the other end is pivoted to the body by a pivot portion, the two side sheets are each provided with a slot, and a first end of the slot is closer to the pivot portion than a second end, the two side sheets each have a hook portion extending from the first end to the second end of the slot to shade a partial opening of the slot; in a locked state, the slots of the two side sheets are respectively aligned with the two longitudinal long grooves, the lock lever is penetratingly disposed in the two longitudinal long grooves and the slots of the two side sheets, the lock lever is located within the scope of the hook portion, so that the lock lever is clamped by the hook portion, and the two side sheets are limited from pivoting relative to the body; when the gravity lock is inverted, the lock lever can be detached from the scope of the hook portion and switched to an unlocked state of the gravity lock, so that the two side sheets can pivot relative to the body,

wherein the side sheet of the gravity lock has a clamping plate and a locking plate, the pivot portion, the slot and the hook portion are all located at the clamping plate, and the side sheet is combined to the first crossbar or the second crossbar by the locking plate, the clamping plate and the locking plate are connected by an inclined portion; the body has an intermediate sheet connecting two opposite sheets, the two longitudinal long grooves are respectively disposed at the two opposite sheets, in the locked state of the gravity lock, the inclined portions of the two side sheets are aligned at rear end faces of the adjacent opposite sheets, and

wherein the first crossbar has two segmented pipes and two end plugs, and the two end plugs are respectively combined with one end of the two segmented pipes, the locking plate of the two side sheets is combined with one segmented pipe and end plug thereof, and the two opposite sheets are combined with another segmented pipe and end plug thereof; in the unfolded state of the table frame, the two end plugs are abutted against each other and form the folding part of the first crossbar.

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